

Super Learner for survival prediction from censored data: definition, simulations, and application

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<https://github.com/foucher-y/talks>

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The problematic of the method choice for survival prediction

- ▶ The prediction of the probability that a subject experienced an event is often of interest.
- ▶ Several regressions can be used for right-censored data.:
 - ▶ Most of the studies use proportional hazard (PH)-based assumption.
 - ▶ Other models such as accelerated failure time (AFT) approaches are not frequent.
- ▶ Regressions are based on assumption such as log-linearity, PH, specific interactions, etc.
- ▶ Machine learning are increasingly used mainly because of their flexibility.
 - ▶ Random survival forests.
 - ▶ Survival neural networks.
 - ▶ Support-vector machines.
 - ▶ Etc.

A super learner (SL) allows us to combine regressions and algorithms.

- ▶ SL is an algorithm that uses cross-validation to estimate the performance of multiple machine learning models.
- ▶ It has been proven to be asymptotically as accurate as the best possible prediction algorithm that is tested.
- ▶ In 2011, Polley and van der Laan proposed a SL for right-censored data.
- ▶ Two R packages were available :
 - ▶ The first one was proposed by Golmakani et al. (2020). It allows us to obtain the linear predictor of a PH regression.
 - ▶ The second one was developed by Westling et al. (2021) with additional learners: several parametric PH models, a generalized additive Cox regression, and a random survival forest.
- ▶ **We aimed to extend these packages to additional learners and loss functions.**

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Notations

- ▶ Consider a subject i in a sample of N independent subjects ($i = 1, \dots, N$).
- ▶ We note the time-to-event by T_i^* .
- ▶ Right-censoring leads the observation of $T_i = \min(T_i^*, C_i)$, where C_i is the censoring time.
- ▶ Let $D_i = \mathbb{I}\{T_i^* \leq C_i\}$ be the event indicator.
- ▶ The survival function at time t for a subject with the characteristics Z_i at baseline is defined by:

$$S(t \mid Z_i) = \mathbb{P}(T_i^* > t \mid Z_i)$$

The SL is estimated by minimizing the cross-validated loss function.

- ▶ $S_m(\cdot)$ is the survival function obtained by the m^{th} learner ($m = 1, \dots, M$).
- ▶ w_m is the corresponding weight with respect to $\sum_1^M w_m = 1$ and $0 \leq w_m \leq 1$.
- ▶ The sample is randomly divided into V cross-validated sub-samples.
- ▶ For each of the folds, one can:
 - ▶ Estimate the M learners from the training subjects.
 - ▶ Tuning parameters should be tuned for each of the V folds.
 - ▶ Predict $\tilde{S}_m(\cdot)$ of the leaving subjects.
- ▶ The weights $\hat{w}_m$ are then obtained by minimizing the loss function, i.e., distance between the observations and the predictions $\tilde{S}_{sl}(t | Z) = \sum_{m=1}^M w_m \tilde{S}_m(t | Z)$.
- ▶ The final SL is obtained by $\hat{S}_{sl}(t | Z) = \sum_{m=1}^M \hat{w}_m \hat{S}_m(t | Z)$, where $\hat{S}_m(t | Z)$ are estimated on the entire sample.

The implemented models without tuning parameters

- ▶ Parametric AFT models (Weibull, Gamma and generalized Gamma distributions).
 - ▶ Parameters are estimated by likelihood maximization by using the "flexsurv" package.
- ▶ Parametric PH models (Exponential or Gompertz distributions).
 - ▶ Parameters are estimated by likelihood maximization by using the "flexsurv" package.
- ▶ Semi-parametric PH models with a non-parametric baseline hazard function estimated by using the Breslow estimator
 - ▶ Parameters are estimated by likelihood maximization by using the "survival" package.
 - ▶ Option for covariates selection by forward AIC-based selection.
 - ▶ Non-parametric baseline hazard function estimated by using the Breslow estimator.

The implemented models/algorithms with tuning parameters

- ▶ Spline-based PH model proposed by Royston and Parmar.
 - ▶ Parameters are estimated by likelihood maximization by using the "flexsurv" package.
- ▶ Penalized PH models (Lasso, Ridge, or Elastic-Net).
 - ▶ Parameters are estimated by penalized likelihood maximization by using the "glmnet" package.
- ▶ Random survival forests.
 - ▶ Maximization of the Log-Rank statistic by using the "randomSRC" package.
- ▶ Survival neural networks.
 - ▶ Maximization of the entropy of a partial logistic regression approach "survivalPLANN" package.
 - ▶ <https://github.com/chupverse/survivalPLANN>

The implemented loss functions.

- ▶ The log-Likelihood.
- ▶ The area under the time-dependant ROC curve up to time t .
- ▶ The Pencina's concordance index at time t .
- ▶ The Uno's concordance index at time t .
- ▶ The Brier Score (BS) for right-censored data and a prediction at time t .
- ▶ The negative binomial log-likelihood (BLL) for a prognostic at time t .
- ▶ The integrated BS and BLL up to the maximum follow-up time.
- ▶ The restricted integrated BS and BLL up to a time t .

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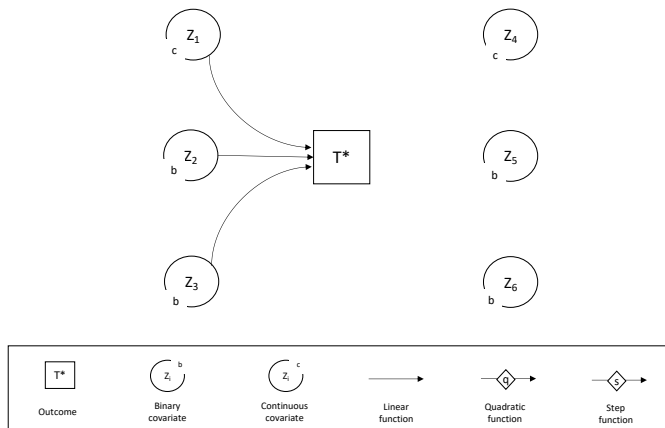
The objectives of the simulations

- ▶ We principally aim to describe the performances of our proposed package.
- ▶ We also compared our SL with the package proposed by Westling (2021).
- ▶ In the SL proposed by Westling we included the 5 possible learners: the PH model with the Breslow estimator, the Exponential PH model, the random survival forest, the Lasso PH model and the PH model with covariates' selection ($p < 0.05$).
- ▶ In our proposed SL, we included the 7 learners: the PH model with the Breslow estimator, the same model with forward selection based on AIC minimization, the Elastic-Net PH model, the random survival forest, the Exponential PH model, the Gamma distribution-based AFT model, and the survival neural network.
- ▶ We used the default grid of the tuning parameters.
- ▶ The SL weights were estimated by minimizing the integrated BS (IBS).

The data generation

- ▶ We generated 1000 data sets for each scenario.
- ▶ The times-to-event were obtained from Weibull distributions and the PH assumption.
- ▶ The censoring times were generated from uniform distributions to obtain a 40% censoring rate.
- ▶ We studied two sample sizes for learning (200 and 500).
- ▶ The validation samples were composed of 500 subjects.
- ▶ We proposed two contrasting simple and complex scenarios.

The design of the simple scenario



Simple scenario: the SL performed as well as semi-parametric approaches

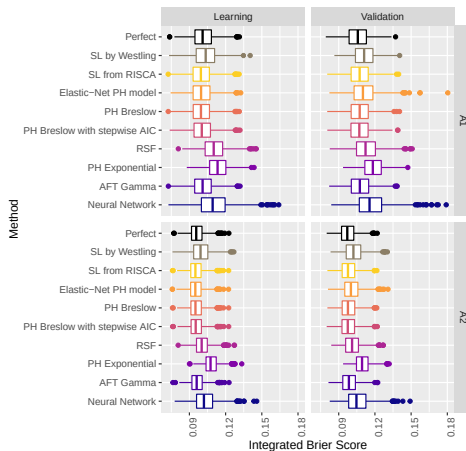
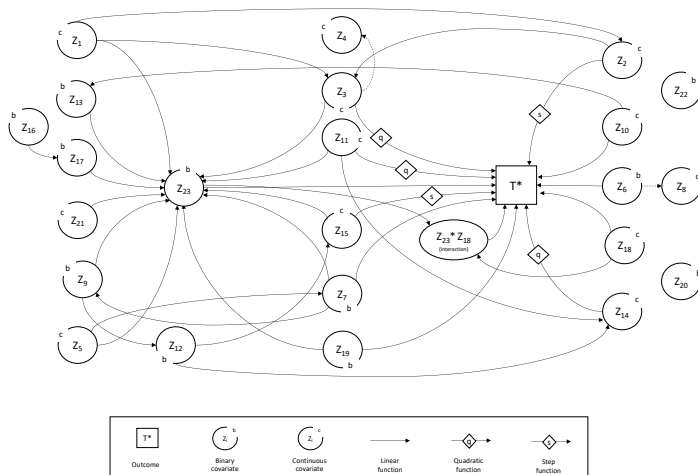


Figure 1: Simulation results in the simple context. $A1-N = 200$ for learning in the two top plots. $A2-N = 500$ for learning in the two bottom plots.

The design of the complex scenario



Complex scenario: the SL performed the best

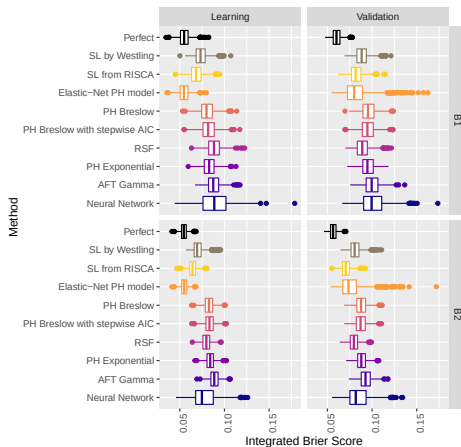


Figure 2: Simulation results in the complex context. A1- $N = 200$ for learning in the two top plots. A2- $N = 500$ for learning in the two bottom plots.

Complex scenario: running times

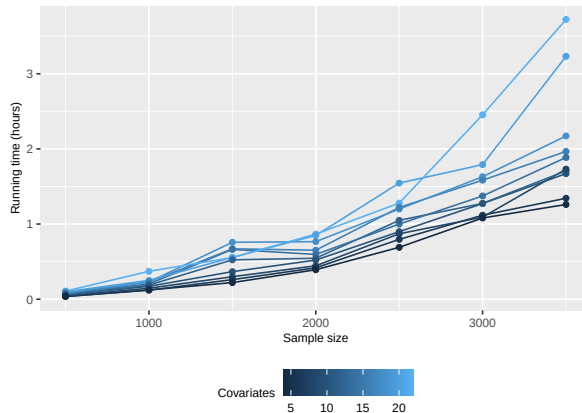


Figure 3: Running times according to the sample size and the number of covariates. Results obtained for a MacBook Pro 2.6 GHz Intel Core i7 6 cores.

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Objectives & data

- ▶ To construct an algorithm for predicting the time to disease progression after the initiation of a first-line treatment (baseline $T^* = 0$).
- ▶ To compare its prognostic capacities up to 2 years with those of the existing Rio score .
- ▶ A data set 1,300 simulated patients from the OFSEP cohort.

```
1 library("survivalSL")
2 data(dataOFSEP); head(dataOFSEP)
3
4 #>   time event age duration period gender relapse edss t1 t2 rio
5 #> 2.1195051 1 34 1113 0 1 1 low 0 0 1
6 #> 0.8160995 1 37 1296 0 1 1 high 1+ 0 1
7 #> 1.9709546 1 33 995 1 1 0 low 0 0 0
8 #> 2.5881311 1 35 858 1 1 0 low 0 0 0
9 #> 1.4726224 1 31 759 1 0 2+ miss 1+ 0 2
10 #> 1.6970962 1 40 1642 1 1 0 high 0 0 0
```

Data management...

```
1 # codage of the predictors
2 dataOFSEP$relapse.1 <- 1*(dataOFSEP$relapse=="1")
3 dataOFSEP$relapse.2 <- 1*(dataOFSEP$relapse=="2+")
4 dataOFSEP$edss.1 <- 1*(dataOFSEP$edss=="low")
5 dataOFSEP$edss.2 <- 1*(dataOFSEP$edss=="high")
6 dataOFSEP$t1.1 <- 1*(dataOFSEP$t1=="0")
7 dataOFSEP$t1.2 <- 1*(dataOFSEP$t1=="1+")
8
9 #two-thirds of the sample to train and other third validation
10 set.seed(117)
11 dataOFSEP$train <- 1*rbinom(n=dim(dataOFSEP)[1], size=1, prob=2/3)
12 dataTRAIN <- dataOFSEP[dataOFSEP$train==1,]
13 dataVALID <- dataOFSEP[dataOFSEP$train==0,]
```

The SL training

- ▶ Learners:
 - ▶ Elastic-Net PH model with non-parametric Breslow hazard function,.
 - ▶ Royston-Parmar Spline-based model.
 - ▶ AFT model with generalized Gamma distribution.
 - ▶ Random survival forest.
- ▶ We used the default grids to estimate the tuning parameters.
- ▶ The weights were estimated to minimize the C-index at 2 years with a 30-fold CV.

```
1 .f <- Surv(time, event) ~ age + duration + period + gender +  
2 relapse.1 + relapse.2 + edss.1 + edss.2 + t1.1 + t1.2  
3  
4 sl1 <- survivalSL(formula=.f, metric="uno_ci", data=dataTRAIN,  
5 methods=c("LIB_COXen", "LIB_PHspline", "LIB_AFTgamma", "LIB_RSF"),  
6 cv=30, optim.local.min=TRUE, show_progress=FALSE, seed=117)
```

The contribution of the learners

```
1
2 print(s11, digits=4)
3
4 #> learners weights
5 #> 1 LIB_COXen 0.1775
6 #> 2 LIB_PHspline 0.2519
7 #> 3 LIB_AFTgamma 0.3039
8 #> 4 LIB_RSF 0.2667
9
10 #> Minimum of the 30-fold CV of the metric uno_ci:0.6851.
```

Your own tuning grid

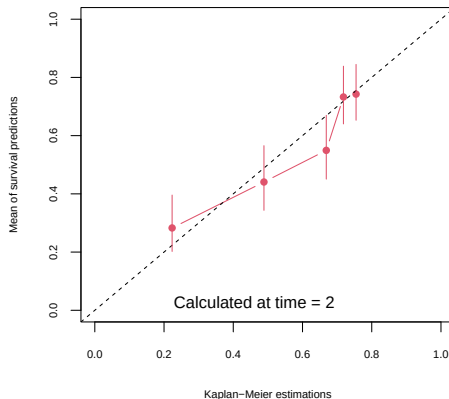
```
1 .tune <- vector("list",4)
2 .tune[[2]] <- list(k=1:6) # number of knots
3
4 sl2 <- survivalSL(formula=.f, metric="uno_ci", data=dataTRAIN,
5 methods=c("LIB_COXen", "LIB_PHSpline", "LIB_AFTgamma", "LIB_RSF"),
6 cv=30, optim.local.min=TRUE, param.tune=.tune, show_progress=FALSE,
7 seed=117)
8
9 print(sl2, digits=4)
10 #> leaners weights
11 #> 1 LIB_COXen 0.2486
12 #> 2 LIB_PHSpline 0.3340
13 #> 3 LIB_AFTgamma 0.1773
14 #> 4 LIB_RSF 0.2401
15
16 #> Minimum of the 30-fold CV of the metric uno_ci:0.684.
```


Predictive metrics from the training and validation samples

```
1 rbind(train = summary(sl2, method="sl", pro.time=2, digits=2)$metrics,  
2 test = summary(sl2, newdata=dataVALID, method="sl", pro.time=2, digits  
   =2)$metrics)  
3  
4 #> p_ci uno_ci auc bs ibs ribs bll ibll ribll ll  
5 #> train 0.7463 0.7444 0.8056 0.1838 0.0813 0.0840 0.5487 NaN NaN  
   -465.7561  
6 #> test 0.6737 0.6713 0.7036 0.2170 0.0924 0.0969 0.6232 NaN NaN  
   -595.7444
```

Calibration plot at 2 years from the validation sample

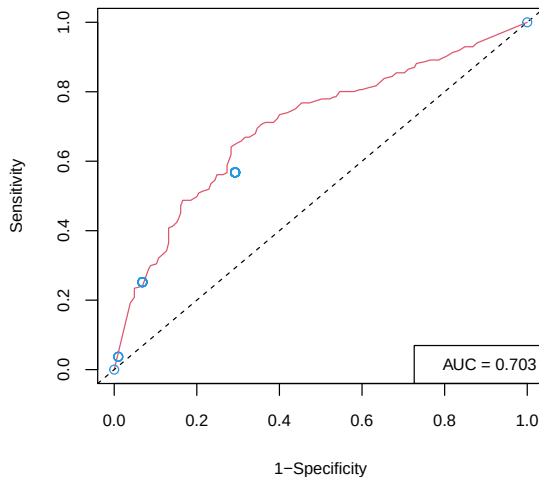
```
1 plot(sl2, newdata=dataVALID, cex.lab=0.70, cex.axis=0.70, n.groups=5,  
    pro.time=2, col=2)
```



Discriminative capacities up to 2 years from the validation sample

```
1 library("RISCA")
2 .pred <- predict(sl2, newdata=dataVALID)
3 dataVALID$sl <- 1 - .pred$predictions$sl[,sum(.pred$times<2)]
4
5 roc.sl <- roc.time(times="time", failures="event", variable="sl",
6                   confounders=~1, data=dataVALID, pro.time=2, precision=seq(0.1, 0.9,
7                                     by=0.01))
8
9 roc.rio <- roc.time(times="time", failures="event", variable="rio",
10                   confounders=~1, data=dataVALID, pro.time=2)
11
12 plot(roc.sl, col=2, type="l", xlab="1-Specificity", ylab="Sensitivity",
13      , cex.lab=0.8, cex.axis=0.8)
14 points(x=1-roc.rio$table$sp, y=roc.rio$table$se, col=4)
15 legend("bottomright", legend=
16 paste("AUC =", round(roc.sl$auc, 3)), cex=0.8 )
```

Discriminative capacities up to 2 years from the validation sample



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- ▶ Compared to the available R-based SL in this context of survival analysis, our proposition allows a larger set of candidate learners and loss functions.
- ▶ The simulation study showed that our proposed functions performed well.
- ▶ The R package is available at:
<https://cran.r-project.org/web/packages/survivalSL/index.html>
- ▶ Bug reports on the Github:
<https://github.com/chupverse/survivalSL/issues>
- ▶ Full open paper for details:
<https://journal.r-project.org/articles/RJ-2024-037>

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